

HW # 17

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Problem 1:-

$$P_3 = \{1, t, t^2, t^3\}$$

$$D(P_3) = \{0, 1, 2t, 3t^2\}$$

$$\text{kernel}(D) = \{1\} \quad \text{since it satisfies } D(P(1)) = 0$$

$$\text{Range}(D) = \{1, 2t, 3t^2\}$$

$$\text{Rank}(D) = 3 \quad \text{since Range is 3 dim}$$

$$\text{Nullity}(D) = 1 \quad \text{since kernel is 1 dim}$$

$$\begin{aligned} \text{Rank}(D) + \text{Nullity}(D) &= \dim(P_3) \\ 3 + 1 &= 4 \end{aligned}$$

Problem 2:-

$$P_1 = \{1, t\}$$

$$S(P_1) = \left\{1, \frac{1}{2}\right\}$$

$$\text{kernel}(S) = \{0\}$$

$$\text{Range}(S) = \left\{1, \frac{1}{2}\right\}$$

$$\text{Rank}(S) = 2$$

$$\text{Nullity}(S) = 0$$

$$\text{Rank} - \text{Nullity} =$$

$$2 + 0 = 2$$

Problem 3:

$$T(A) = A^T \Rightarrow A^T = 0 \text{ iff } A = 0$$

$$\text{kernel}(T) = \{0\}$$

$$\text{Range}(T) = 4$$

$$\text{Rank}(T) = 4$$

$$\text{Nullity}(T) = 0$$

since 2×2 matrices
dimension is 4
& all matrices have
unique transpose.

$$4 + 0 = 4$$

Problem 4:

$$V = \text{Span}\{e^x, xe^x, x^2e^x\}$$

Both V & $\mathbb{R}^3$ are 3 dimensional

$$T: V \rightarrow \mathbb{R}^3 \quad T(e^x) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad T(xe^x) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad T(x^2e^x) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\rightarrow f(x) = ae^x + bxe^x + cx^2e^x$$

$$\begin{aligned} T(f(x)) &= aT(e^x) + bT(xe^x) + cT(x^2e^x) \\ &= \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in \mathbb{R}^3 \end{aligned}$$

$$\begin{aligned} \rightarrow T(f(x) + g(x)) &= T((a_1 + a_2)e^x + (b_1 + b_2)xe^x + (c_1 + c_2)x^2e^x) \\ &= \begin{bmatrix} a_1 + a_2 \\ b_1 + b_2 \\ c_1 + c_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} + \begin{bmatrix} a_2 \\ b_2 \\ c_2 \end{bmatrix} \end{aligned}$$

$$\rightarrow T(\alpha f(x)) = \alpha a T(e^x) + \alpha b T(xe^x) + \alpha c T(x^2e^x)$$

$$= \begin{bmatrix} \alpha a \\ \alpha b \\ \alpha c \end{bmatrix} = \alpha \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Therefore,

$$V \cong \mathbb{R}^3$$